

ECEN 607: Advanced Analog Circuit Tech

ASSN 1:REPORT

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2 Use the technology you are using in the lab and characterize a single N-type transistor employing

$W=4.8\mu\text{m}$ and $L=0.18\mu\text{m}$. Use the IBM 0.18 technology you are using in the laboratory; please check it with your TA.

i) Set the drain-source voltage at 1V, and sweep the gate-source voltage from 0 up to 1.5V. Plot I_{ds} , G_m , dI_{ds}/dV_{gs} , and d^2I_{ds}/dV_{gs}^2 versus gate-source voltage. Have enough points (>20) to have smooth curves.

ii) For same conditions as i), plot C_{gs} and G_m/C_{gs} versus $V_{gs}-V_T$ (overdrive voltage); the parameter represents the transistor's f_t , which is quite relevant when designing fast circuits; G_m/C_{gs} is also known as the transistor's f_t .

iii) Set the gate voltage such that the overdrive voltage $V_{gs}-V_T=200\text{mV}$, and sweep the drain voltage in the range 0-1.8V. Extract the transistor's output resistance and plot R_{ds} , dR_{ds}/dV_{ds} , and d^2R_{ds}/dV_{ds}^2 versus drain-source voltage. Notice that the derivatives of R_{ds} represent output resistor non-linearities. Certainly, R_{ds} is a second order effect and very non-linear, though!

iv) Repeat iii) for temperatures of -50, 27 and 100 degrees (Celsius). Make a table and compare I_d , G_m and R_{ds} for the 3 cases. For all these simulations keep constant the gate-source voltage; evidently V_{dsat} will not be constant due to V_T variations.

v) Simulate the transistor and report the input and output referred noise density. Include a screen shot showing the noise density in the frequency range of 1Hz up to 10MHz; use $\log_{10}$ scale in the x-axis and y-axis to visualize the flicker noise level (voltage) and the thermal noise level (current). Check these values with the theoretical values.

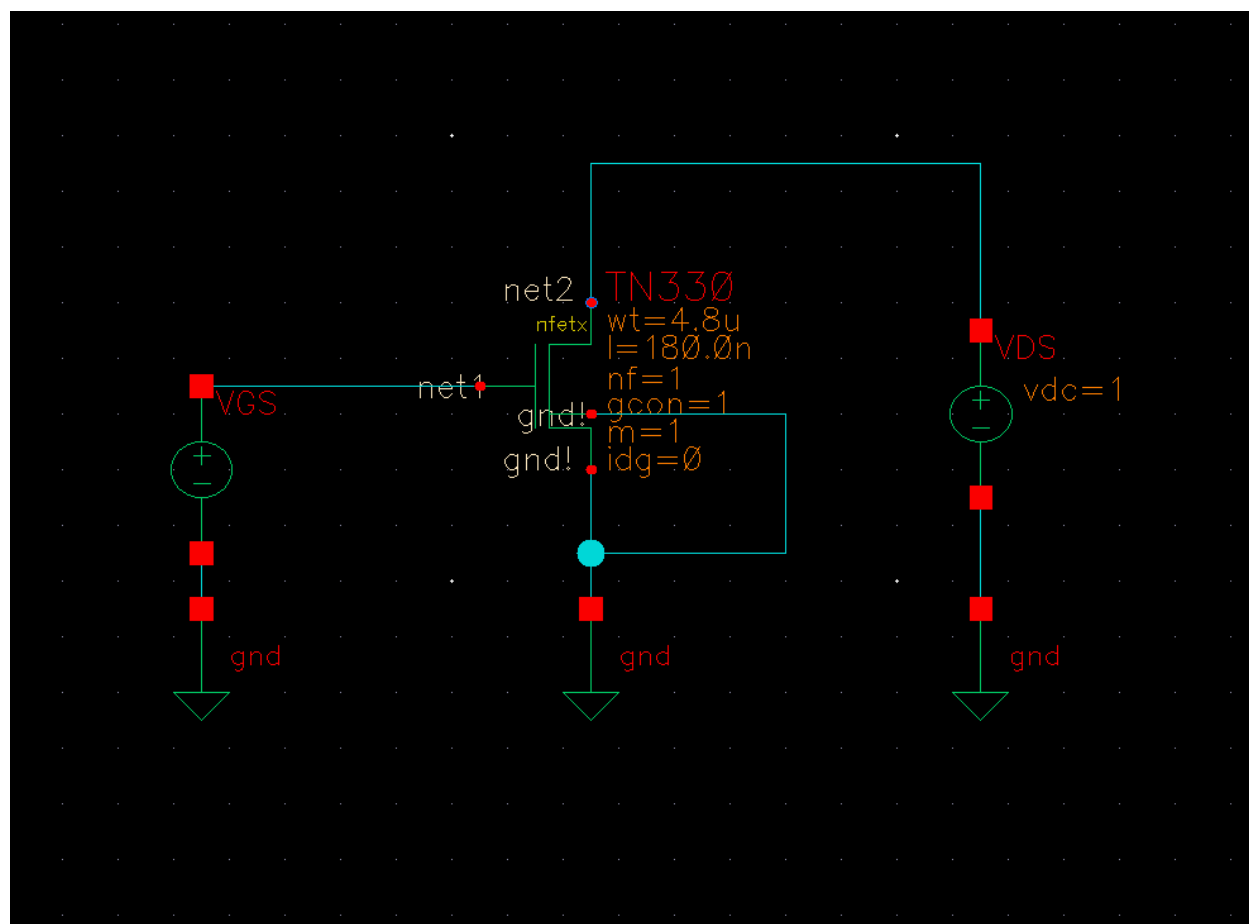
The easiest way to do this is by biasing the gate voltage with a DC voltage source such that $V_{gs}-V_T=200\text{mV}$ in series with an AC voltage source.

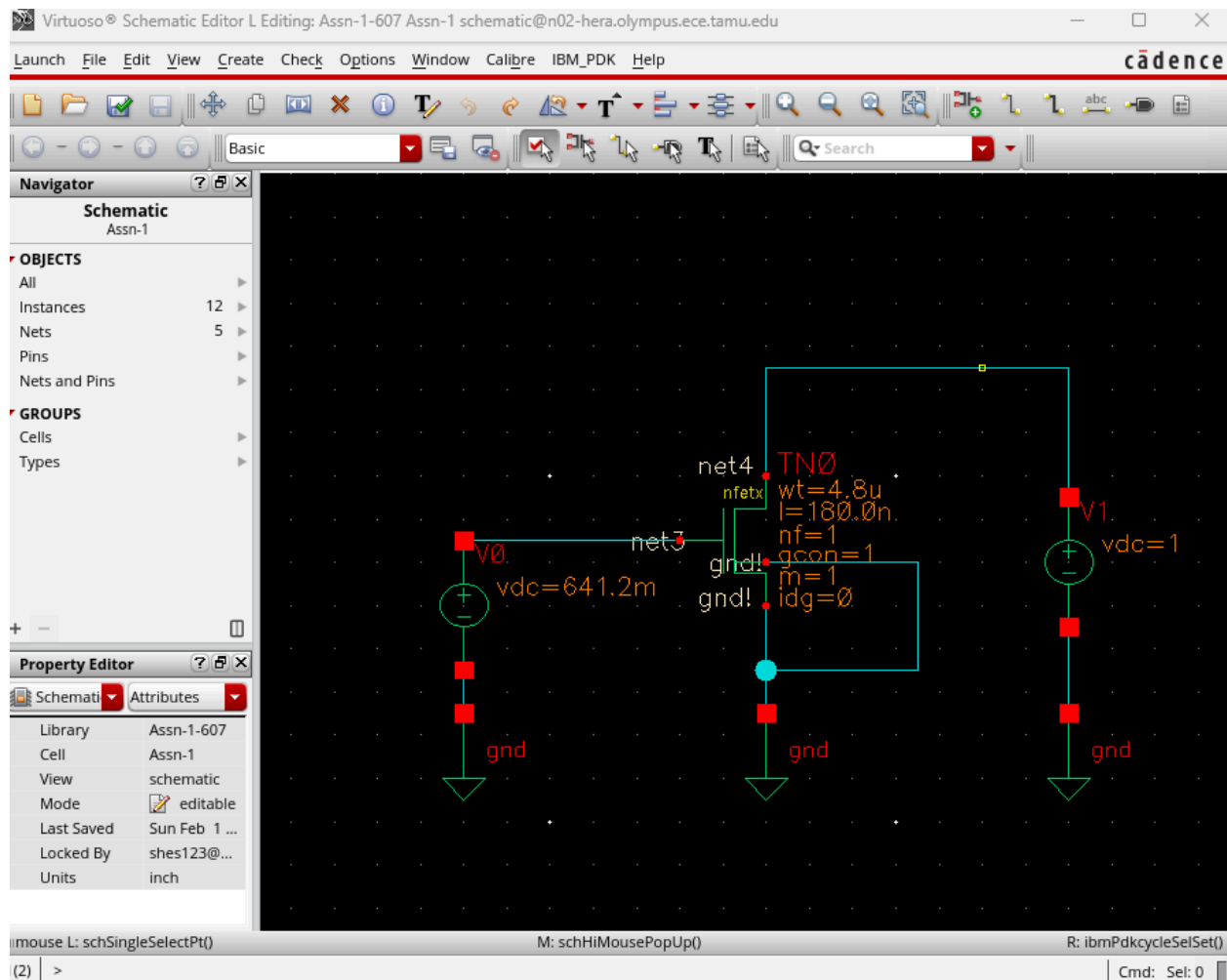
Connect at amplifier's drain terminal a power supply such that V_{DS} is set at 1V. The DC voltage source operates as short circuit for AC analysis and current can be measured.

1, TABULATION OF VALUES:

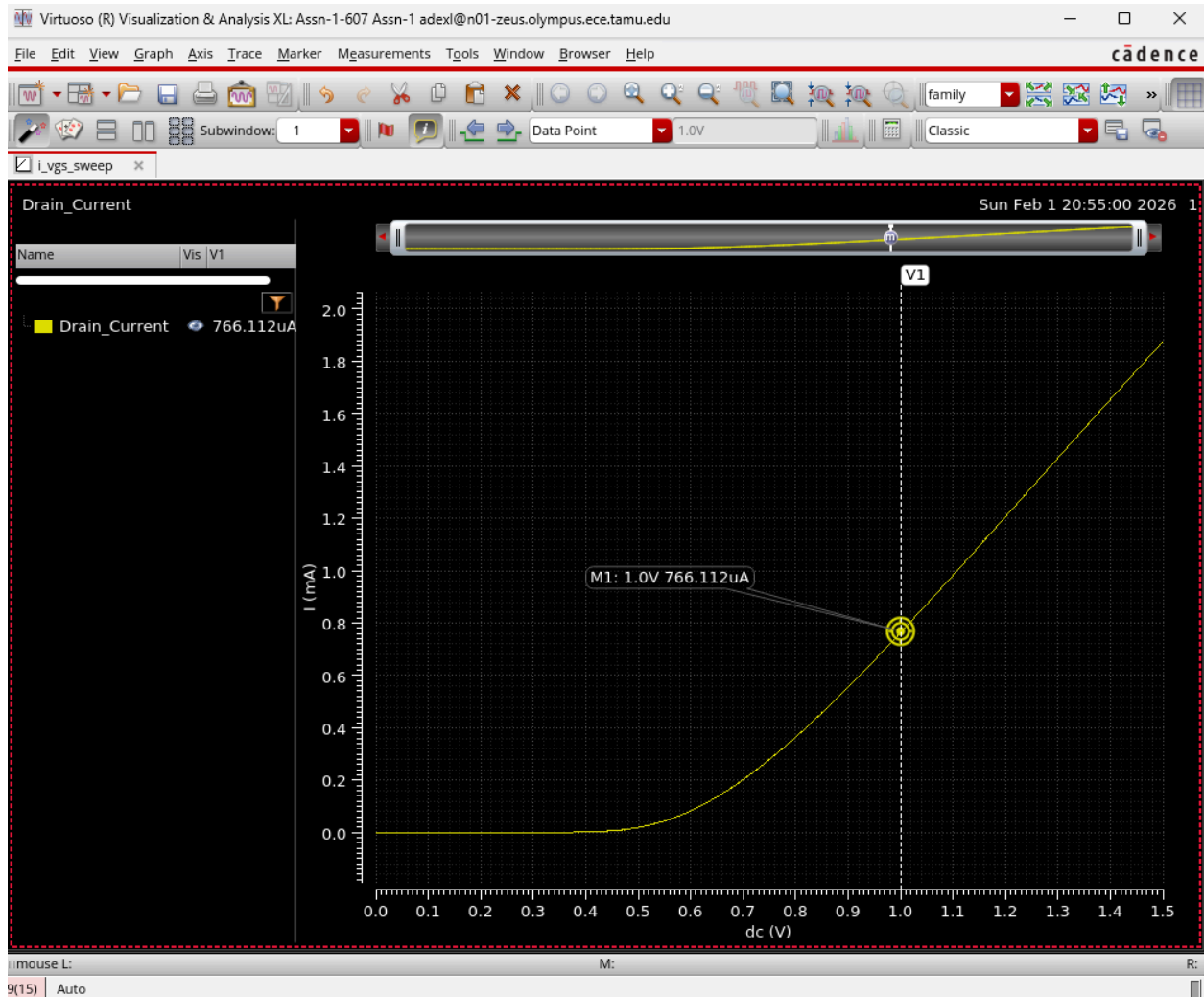
TEMPERATURE	ID CURRENT
-50	107u
27	127u
100	145u

2, SCHEMATIC SCREENSHOTS:

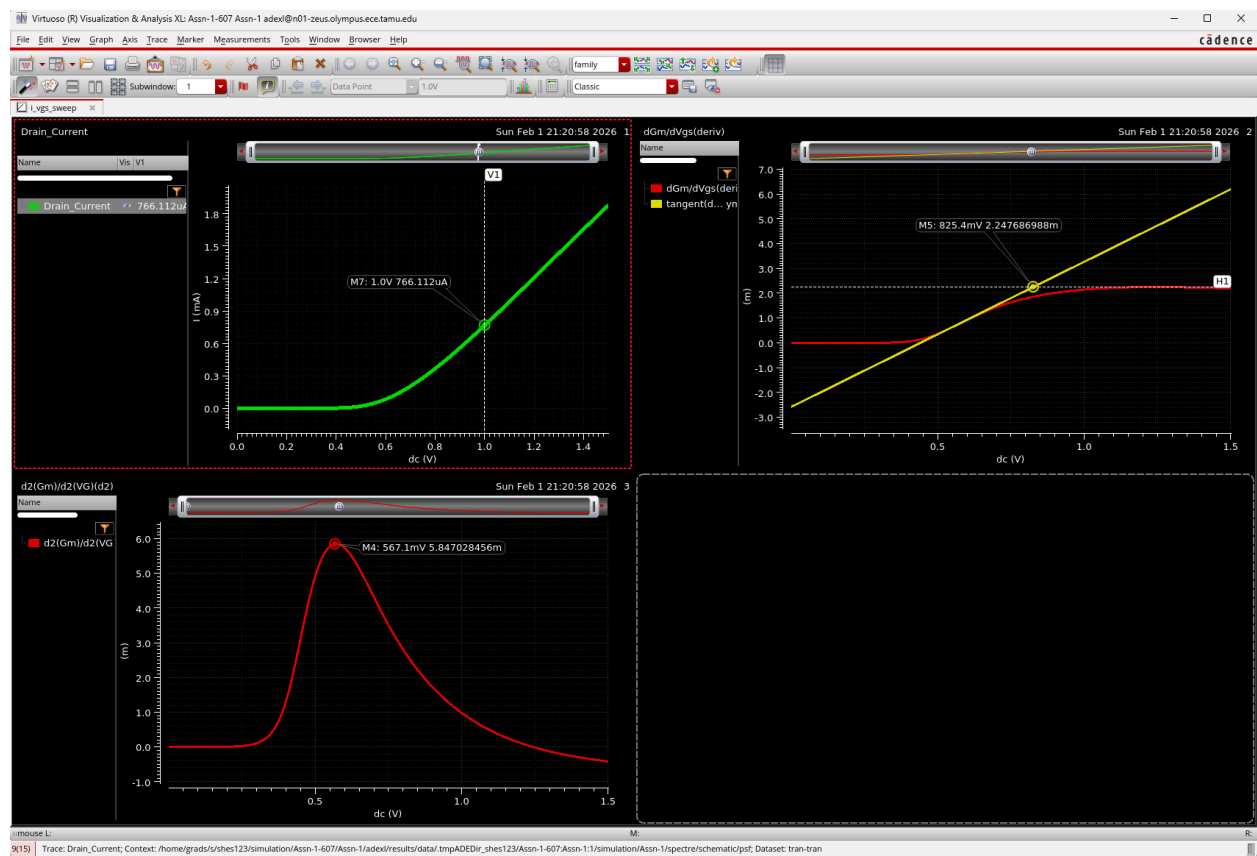




3.SIMULATION SCREENSHOTS:



i) Set the drain-source voltage at 1V, and sweep the gate-source voltage from 0 up to 1.5V. Plot I_{ds} , G_m , $\frac{dI_{ds}}{dV_{gs}}$, and $\frac{d^2I_{ds}}{dV_{gs}^2}$ versus gate-source voltage. Have enough points (>20) to have smooth curves.



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Launch File Create Tools Options Run EAD Parasitics/LDE Window Calibre Help

No Parasitics/LDE Single Run, Sweeps and Corners

Data View

- Tests
 - ☒ i_vgs_sweep
- Global Variables
- Parameters
- Corners
- Documents
- Setup States
- Reliability Analyses
- Checks/Asserts

Data History

Run Summary

1 Test ☒ Nominal Corner

☒ 1 Point Sweep ☒ 0 Corner

History Item	Status
Interactive.1	finished
Interactive.0	finished

Outputs Setup **Results**

Detail

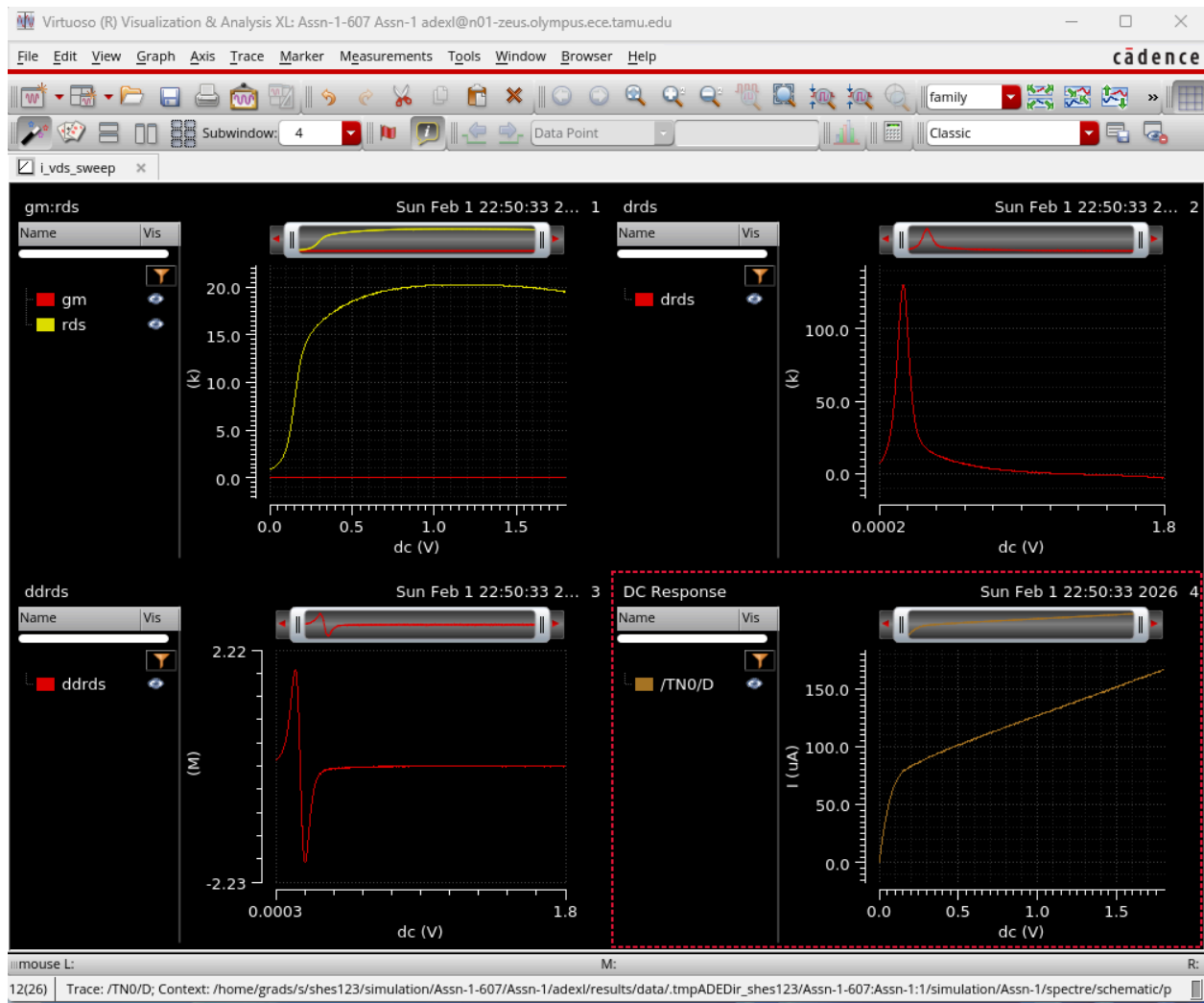
Test	Output	Nominal	Spec	Weight	Pass/Fail
i_vgs_sweep	Drain_Current				
i_vgs_sweep	/TN330/D				
i_vgs_sweep	dGm/dVgs(deriv)				
i_vgs_sweep	d2(Gm)/d2(VG)...				
i_vgs_sweep	ymaxdd	5.85m			
i_vgs_sweep	xvaluedd	571.4m			
i_vgs_sweep	yvalued	761.8u			
i_vgs_sweep	tangent				
i_vgs_sweep	ymax_d	2.248m			
i_vgs_sweep	vtn	441.2m			
i_vgs_sweep	vt+(1/2theta)	825.4m			

Interactive.0 Interactive.1

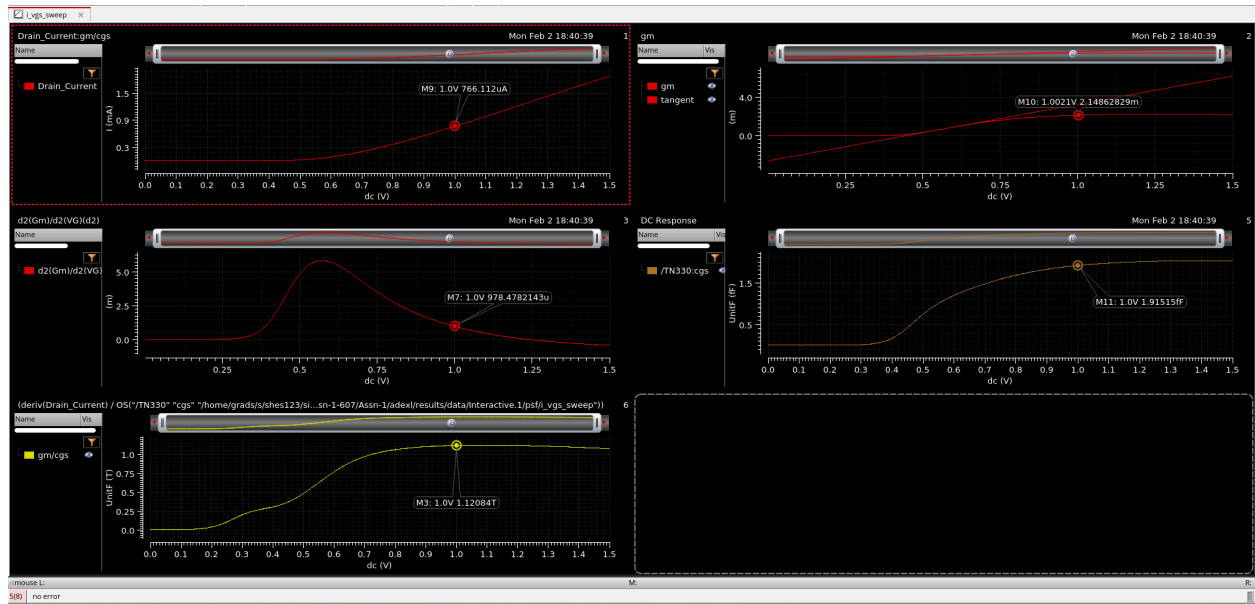
mouse L: M: R:

5(8)

rds:



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for temperatures of -50, 27 and 100 degrees (Celsius). Make a table and compare I_d , G_m and R_{ds} for the 3 cases. For all these simulations keep constant the gate-source voltage;
1,-50

ADE XL Test Editor - i_vds_sweep@n02-hera.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

cadence

Design Variables

Name	Value
p5vonly	0

Analyses

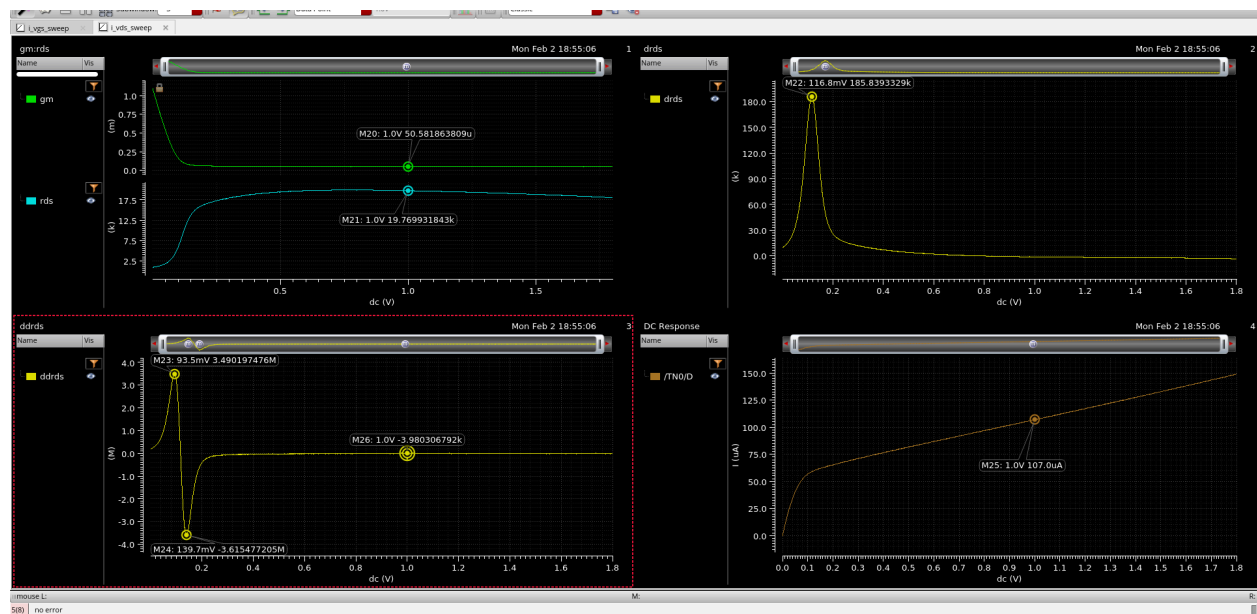
Type	Enable	Arguments
1 dc	☒	t 0 1.8 100u Linear Step Size Start-Stop /V1

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 drain_current	107u	☒	☐	
2 id		☒	☒	yes
3 gm	wave	☒	☐	
4 rds	wave	☒	☐	
5 drds	wave	☒	☐	
6 ddrds	wave	☒	☐	

> Results in ...r_shes123/i_vds_sweep/simulation

3(4) Debug Test | Status: Ready | T=-50.0 C | Simulator: spectre aps | State: i_vds_sweep_active



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Session Setup Analyses Variables Outputs Simulation Results Tools Help

27

Design Variables

Name	Value
1 p5vonly	0

Analyses

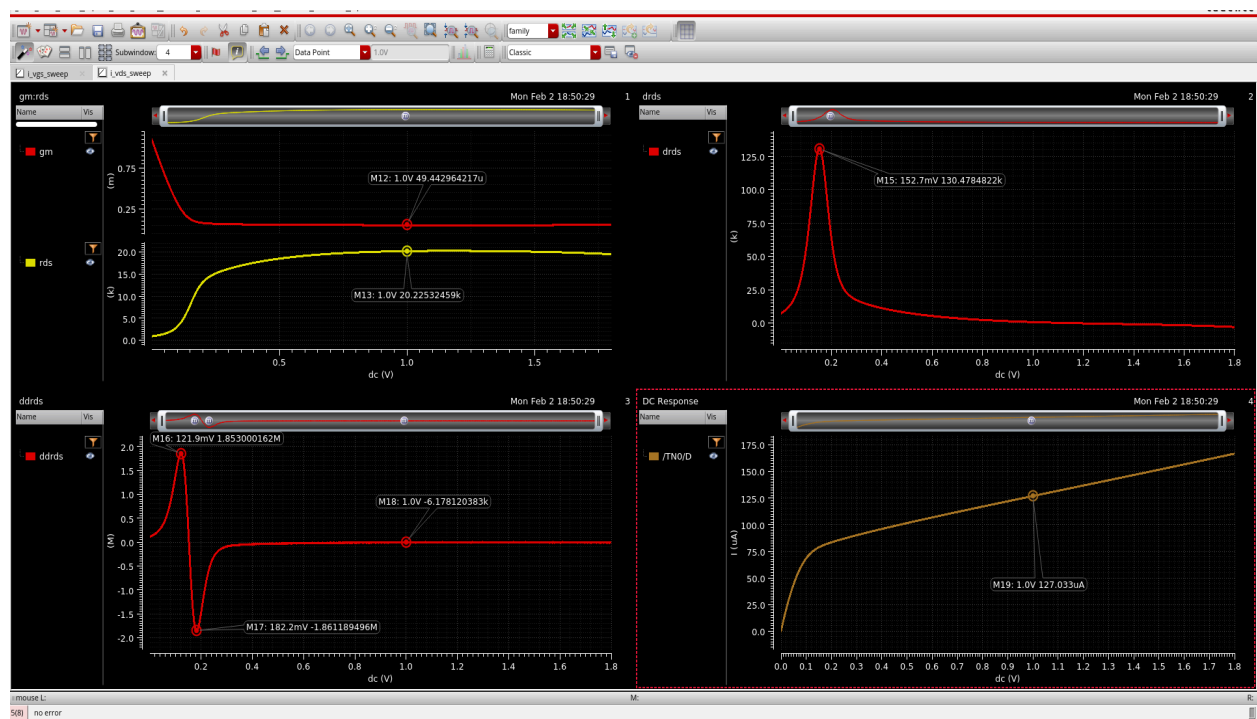
Type	Enable	Arguments
1 dc	☒	t 0 1.8 100u Linear Step Size Start-Stop /V1

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 drain_current	127u	☒	☐	
2 id		☒	☒	yes
3 gm		☒	☐	
4 rds		☒	☐	
5 drds		☒	☐	
6 ddrds		☒	☐	

> Select on Schematic Instances for operating po

3(4) Temperature | Status: Ready | T=27 C | Simulator: spectre aps | State: i_vds_sweep_active



3.100

ADE XL Test Editor - i_vds_sweep@n02-hera.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

100.0

Design Variables

Name	Value
p5vonly	0

Analyses

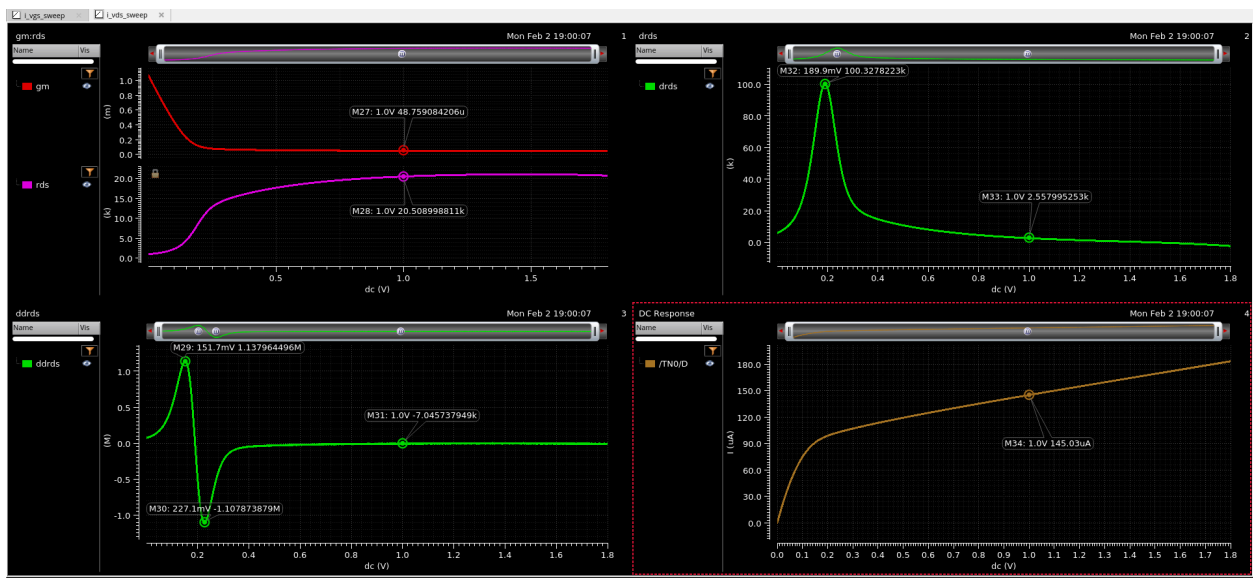
Type	Enable	Arguments
1 dc	☒	t 0 1.8 100u Linear Step Size Start-Stop /V1

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 drain_current	145u	☒	☐	
2 id		☒	☒	yes
3 gm	wave	☒	☐	
4 rds	wave	☒	☐	
5 drds	wave	☒	☐	
6 ddrds	wave	☒	☐	

> Results in ...r_shes123/i_vds_sweep/simulation

3(4) Debug Test Status: Ready T=100.0 C Simulator: spectre aps State: i_vds_sweep_active



v) Simulate the transistor and report the input and output referred noise density. Include a screen shot showing the noise density in the frequency range of 1Hz up to 10MHz; use log10 scale in the x-axis and y-axis to visualize the flicker noise level (voltage) and the thermal noise level (current). Check these values with the theoretical values.

The easiest way to do this is by biasing the gate voltage with a DC voltage source such that $V_{gs}-V_T=200\text{mV}$ in series with an AC voltage source.

Connect at amplifier's drain terminal a power supply such that V_{DS} is set at 1V. The DC voltage source operates as short circuit for AC analysis and current can be measured.

PLOTS SHOWING THE THERMAL AND FLICKER NOISE:

